



Fig. 1: Broad scope of agri-electronics

key to increasing the contribution of ICT research in agricultural development.

National status

Agri-electronics activities at the national level may be divided into two major categories: Electronics and instrumentation for agriculture, and ICT for agriculture.

While very few national R&D laboratories are involved in the field of electronics and instrumentation, government agencies, NGOs and ICT companies in the private sector are involved in developing and deploying e-solutions mainly focused on information-dissemination services. Such ICT services include advisory information provision over the Internet or mobile devices to the farmers on weather forecasts, price information, etc.

IT and electronics in agriculture.

Table I shows the R&D labs in India that are engaged in agri-electronics research to develop novel and smart instruments.

Agri-information dissemination systems. In India, there are 46 Web portals and 39 IT-based Web-enabled applications already operational. All these portals and applications provide information transaction services to the

government departments, agro-marketers at all levels, farmers and all the stakeholders in agricultural domain.

Though a number of these Web-based portals have very friendly user-interfaces and presentations in vernacular languages, these are not yet used regularly by the small and medium farmers due to a number of reasons including lack of infrastructure, lack of IT literacy and inadequate content. Of course, the portals find good use in

government-to-government transactions.

A comprehensive farmers' portal is under development by the Ministry of Agriculture & Cooperation, government of India. Mobile governance is also making its way to the e-governance arena, and a number of solutions using mobile-based agro-advisory services are being developed by a number of companies. A brief synopsis of such e-agriculture initiatives is given in Table II.

The following conclusions can be drawn from the above discussion:

1. In India, more stress is in deploying e-agriculture solutions over the Internet or mobile devices by multi-utility Web portals.

2. No focus is directed towards development of smarter electronics and instrumentation systems in India.

3. Even the deployed ICT projects in India provide very few success stories, and there is no widespread outcome and impact at the grassroots level.

In view of the above, there exists an emergent need for focused and integrated endeavour towards development of smart devices for the agricultural sector and coupling of the smart devices to the intelligent software through advanced telecommunication (Internet or mobile devices).

A few innovative solutions have been developed for quality measurement, monitoring and management of Indian crops. These innovations are detailed in the box.

TABLE I
R&D Labs Engaged in Agri-electronics Research

Name of the Institute	Brief details of activities
C-DAC, Kolkata	Tea withering monitor, jute auto-leveller, integrated automation of tea processes at Tocklai, Jorhat, electronic nose, electronic vision and electronic tongue
Central Electronics Engineering Research Institute (CEERI), Pilani	Smart pond management system for fresh water aquaculture, automation of RO plants, sensors, controlled-atmosphere storage house and sugar processing automation
Central Scientific Instruments Organisation (CSIO), Chandigarh	Foodgrain sorter, sensors and sensing systems, and optical instruments
Department of Electronics and Accreditation of Computer Classes (DOEACC), Aurangabad	Microprocessor-based seed de-husking system, environmental sensors and automatic root dendrometer